

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent process started.\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("\n--- CHILD PROCESS ---\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());
        printf("Child is RUNNING.\n");

        printf("Child is going to WAIT for 10 seconds...\n");
        sleep(10);

        printf("Child is RUNNING again after waiting.\n");

        printf("Child terminating...\n");
        exit(0);
    }
    else
    {
        // Parent process
        printf("\n--- PARENT PROCESS ---\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
        printf("Parent is RUNNING.\n");

        printf("Parent is WAITING for child to finish...\n");

        wait(NULL);

        printf("Child has terminated.\n");
        printf("Parent is RUNNING again.\n");

        printf("Parent terminating...\n");
        return 0;
    }
}

```

Output

```
Parent process started.  
Parent PID : 2045  
Parent PPID : 1722  
  
--- PARENT PROCESS ---  
Parent PID : 2045  
Child PID : 2046  
Parent is RUNNING.  
Parent is WAITING for child to finish...  
--- CHILD PROCESS ---  
Child PID : 2046  
Child PPID : 2045  
Child is RUNNING.  
Child is going to WAIT for 10 seconds...  
Child is RUNNING again after waiting.  
Child terminating...  
Child has terminated.  
Parent is RUNNING again.  
Parent terminating...  
ubuntu@ubuntu:~/OSLab/Practical$
```